

Bacteriophage biocontrol in animals and meat products

Summary Since their discovery almost a century ago, bacterial viruses (bacteriophages or 'phages') have been used to prevent and treat a multitude of bacterial infections (phage therapy: PT). In addition, they have been the basis for many advances in genetics and biochemistry. Phage therapy was performed on human subjects in the United States, Europe and Asia in the few decades following their discovery. However, Western countries largely abandoned PT in favour of antibiotics in the 1940s. The relatively recent renaissance of PT in the West can be attributed partly to the increasing prevalence of antibiotic resistance in human and animal pathogens. However, the stringent controls on human trials now required in the United States and Europe have led to a greater number of domestic animal and agricultural applications as an alternative to PT in man. This trend is set to continue, at least in the short term, with recent approval from the Food and Drug Administration allowing commercial phage treatments to be used in human food in the USA. Nevertheless, despite these significant milestones and the growing number of successful PT trials, significant obstacles remain to their widespread use in animals, food and ultimately medicine in many parts of the world. This review will provide a brief overview of the history of PT in the West and will summarize some of the key findings of phage biocontrol studies in animals and meat products.